Exercise 1: Tensor Calculus

In this exercise, we will go through different aspects of tensor calculus that will enable you to follow and understand the upcoming lectures.

Task 1.1: Einstein Summation Convention

Expand the following expressions into their full forms

a) $v_k \boldsymbol{e}_k$,

b) $A_{ij} B_{ji}$,

c) $b_i = A_{ij} v_j$.

Task 1.2: Rules for Indices

Fill in the missing indices of the tensor $\boldsymbol{A}$

a) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mm}$,

b) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mr}$,

c) $A \quad B_{kr} C_{jks} \delta_{ji} + E_{ml} \delta_{ur} F_{lu} = G_{mr}$

Task 1.3: Transpose of a Tensor

Given the Tensor $\boldsymbol{T} = T_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j$, determine the coefficients of its transpose $\boldsymbol{T}^T = T^T_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j$ using the following definition

$$\boldsymbol{w} \cdot \boldsymbol{T} \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{T}^T \boldsymbol{w} \tag{1.1}$$

Task 1.4: Different Notations

Write the following expressions in symbolic and matrix notation

a) $v_i = A_{ik} w_k$,

b) $C_{ik} = A_{ij} B_{jk}$,

c) $A_{ij} B_{ji}$,

d) $u_j = A_{ik} B_{ij} v_k$,

e) $A_{ij} - \frac{1}{3} \delta_{ln} \delta_{ij} \delta_{km} \delta_{kl} A_{mn}$.

Task 1.5: Symmetric Tensors

A Tensor $\boldsymbol{S}$ is symmetric if it is equal to its transpose $\boldsymbol{S}^T$, i. e. $\boldsymbol{S} = \boldsymbol{S}^T$. For an antisymmetric/skew-symmetric Tensor $\boldsymbol{A}$ however, $\boldsymbol{A} = -\boldsymbol{A}^T$ holds true.

Each Tensor $\boldsymbol{T}$ can be additively decomposed into its symmetric and skew symmetric parts as follows

$$\boldsymbol{T} = \mathrm{sym}(\boldsymbol{T}) + \mathrm{skew}(\boldsymbol{T}), \tag{1.2}$$

with

$$\mathrm{sym}(\boldsymbol{T}) = \tfrac{1}{2}\left(\boldsymbol{T} + \boldsymbol{T}^T\right) \qquad \text{und} \qquad \mathrm{skew}(\boldsymbol{T}) = \tfrac{1}{2}\left(\boldsymbol{T} - \boldsymbol{T}^T\right). \tag{1.3}$$

Many tensors in mechanics, such as the stress tensor $\boldsymbol{\sigma}$ and the strain tensor $\boldsymbol{\varepsilon}$, are symmetric, leading to a lot of simplifications.

Given a symmetric second order tensor $\boldsymbol{S}$ and a general second order tensor $\boldsymbol{T}$, prove that

$$\boldsymbol{S} : \boldsymbol{T} = \boldsymbol{S} : \mathrm{sym}(\boldsymbol{T}). \tag{1.4}$$

Task 1.6: Volumetric–Isochoric Decomposition and some Derivatives

The strain tensor $\boldsymbol{\varepsilon}$ can be decomposed into volumetric and deviatoric parts. The volumetric part is defined by the spherical (hydrostatic) tensor

$$\mathrm{sph}[\boldsymbol{\varepsilon}] = \tfrac{1}{3}\,\mathrm{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I}, \tag{1.5}$$

which is the second order identity tensor multiplied by a scalar, and appears, for instance, in case of hydrostatic pressure.

The trace of the strain tensor $\boldsymbol{\varepsilon}$, which is known as the volume strain $\varepsilon_{\mathrm{V}} = \mathrm{tr}[\boldsymbol{\varepsilon}] = \varepsilon_{kk}$ is a measure of volume change during deformation.

The volume strain can be obtained from the spherical tensor ($\mathrm{tr}[\boldsymbol{\varepsilon}] = \mathrm{tr}[\mathrm{sph}\,\boldsymbol{\varepsilon}]$), and hence $\mathrm{sph}[\boldsymbol{\varepsilon}]$ is called the volumetric part of the strain tensor.

On the other hand, the deviatoric part of the strain tensor

$$\mathrm{dev}[\boldsymbol{\varepsilon}] = \boldsymbol{\varepsilon} - \tfrac{1}{3}\,\mathrm{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I}, \tag{1.6}$$

is traceless, and it describes the volume preserving (isochoric) part of the deformation.

Evaluate the following derivatives in preparation for the next task

a) $\dfrac{\partial\,\boldsymbol{\varepsilon}}{\partial\,\boldsymbol{\varepsilon}}$, b) $\dfrac{\partial\,\mathrm{tr}[\boldsymbol{\varepsilon}]}{\partial\,\boldsymbol{\varepsilon}}$, c) $\dfrac{\partial\,\mathrm{dev}[\boldsymbol{\varepsilon}]}{\partial\,\boldsymbol{\varepsilon}}$, d) $\dfrac{\partial\,||\,\mathrm{dev}\,\boldsymbol{\varepsilon}||}{\partial\,\boldsymbol{\varepsilon}}$.

Task 1.7: Matrix and Voigt Notations of Tensors

Whereas second order tensors such as the stress tensor $\boldsymbol{\sigma}$ and the strain tensor $\boldsymbol{\varepsilon}$, can be directly represented as $3\,\mathrm{x}\,3$ matrices, it is not directly achievable for higher order tensors. The relationship between $\boldsymbol{\sigma}$ und $\boldsymbol{\varepsilon}$, is given by

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}, \tag{1.7}$$

where $\mathbb{C}$ is a fourth order tensor with $3^4 = 81$ coefficients. In order to represent Equation (1.7) in matrix notation, the components of $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ can be arranged into two $9\,\mathrm{x}\,1$ and the components of $\mathbb{C}$ in a $9\,\mathrm{x}\,9$ matrix:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{21} \\ \sigma_{32} \\ \sigma_{31} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113} & \mathbb{C}_{1121} & \mathbb{C}_{1132} & \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213} & \mathbb{C}_{2221} & \mathbb{C}_{2232} & \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313} & \mathbb{C}_{3321} & \mathbb{C}_{3332} & \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213} & \mathbb{C}_{1221} & \mathbb{C}_{1232} & \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313} & \mathbb{C}_{2321} & \mathbb{C}_{2332} & \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313} & \mathbb{C}_{1321} & \mathbb{C}_{1332} & \mathbb{C}_{1331} \\ \mathbb{C}_{2111} & \mathbb{C}_{2122} & \mathbb{C}_{2133} & \mathbb{C}_{2112} & \mathbb{C}_{2123} & \mathbb{C}_{2113} & \mathbb{C}_{2121} & \mathbb{C}_{2132} & \mathbb{C}_{2131} \\ \mathbb{C}_{3211} & \mathbb{C}_{3222} & \mathbb{C}_{3233} & \mathbb{C}_{3212} & \mathbb{C}_{3223} & \mathbb{C}_{3213} & \mathbb{C}_{3221} & \mathbb{C}_{3232} & \mathbb{C}_{3231} \\ \mathbb{C}_{3111} & \mathbb{C}_{3122} & \mathbb{C}_{3133} & \mathbb{C}_{3112} & \mathbb{C}_{3123} & \mathbb{C}_{3113} & \mathbb{C}_{3121} & \mathbb{C}_{3132} & \mathbb{C}_{3131} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{32} \\ \varepsilon_{31} \end{bmatrix}. \tag{1.8}$$

Due to symmetry, the number of independent coefficients of a tensor can be reduced. For instance, a second order symmetric tensor can be uniquely defined using 6 coefficients instead of 9. This is valid for both stress $\boldsymbol{\sigma}$ and strain $\boldsymbol{\varepsilon}$ tensors. The tensor $\mathbb{C}$ also has the following symmetries

$$\mathbb{C}_{ijkl} = \mathbb{C}_{jikl} = \mathbb{C}_{ijlk}, \tag{1.9}$$

that reduce the number of its independent coefficients to 36. Additionally, its major symmetry

$$\mathbb{C}_{ijkl} = \mathbb{C}_{klij}, \tag{1.10}$$

further reduces the number of independent coefficients to 21. The aforementioned symmetries are exploited to represent the fourth order tensor $\mathbb{C}$ as a matrix using Voigt notation. Therefore, the two second order tensors $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ can be written as column vectors with 6 components, and the fourth order tensor $\mathbb{C}$ as a $6\,\mathrm{x}\,6$ matrix.

a) Write the Equation (1.8) in Voigt notation.

b) Write down the projection tensor $\mathbb{P} = \mathbb{I} - \frac{1}{3}\,\boldsymbol{I} \otimes \boldsymbol{I}$ as a $6\,\mathrm{x}\,6$ matrix using the Voigt notation, such that the deviatoric part of $\boldsymbol{\varepsilon}$ can be obtained by multiplying the aforementioned matrix with $\underline{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{22} & \varepsilon_{33} & 2\,\varepsilon_{12} & 2\,\varepsilon_{23} & 2\,\varepsilon_{13} \end{bmatrix}^T$.

Exercise 1: Tensor Calculus

Task 1.1: Einstein Summation Convention

Indices that occur twice in a product term must be summed from 1 to 3, so the following applies

a) $v_k \boldsymbol{e}_k = v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 + v_3 \boldsymbol{e}_3$,

b) $$\begin{aligned} A_{ij} B_{ji} &= A_{1j} B_{j1} + A_{2j} B_{j2} + A_{3j} B_{j3} \\ &= A_{11} B_{11} + A_{12} B_{21} + A_{13} B_{31} \\ &+ A_{21} B_{12} + A_{22} B_{22} + A_{23} B_{32} \\ &+ A_{31} B_{13} + A_{32} B_{23} + A_{33} B_{33}, \end{aligned}$$

c) $b_i = A_{ij} v_j \quad \Leftrightarrow \quad b_i = A_{i1} v_1 + A_{i2} v_2 + A_{i3} v_3$.

This equation describes the three cases

$$\begin{aligned} i = 1: &\quad b_1 = A_{11} v_1 + A_{12} v_2 + A_{13} v_3, \\ i = 2: &\quad b_2 = A_{21} v_1 + A_{22} v_2 + A_{23} v_3, \\ i = 3: &\quad b_3 = A_{31} v_1 + A_{32} v_2 + A_{33} v_3. \end{aligned}$$

Task 1.2: Rules for Indices

If an index is a free index, then it appears *only* once in each product term. However, if an index is a summation index, it *must* appear twice in a product term.

a) The indices j, k and m each appear twice within a product term and are therefore summation indices. As neither i, r nor s appear on the right-hand side, they cannot be free indices. They must therefore be added to the symbol A: $A_{irs} \delta_{ij} B_{kr} C_{jks} = D_{mm}$

b) The indices j and k are again summation indices. Since m and r each appear only once in the right-hand expression, they are free indices that must also appear once in the left-hand expression. The index m must therefore be added to A. The indices i and s should also be added, as they cannot be free indices and must therefore appear twice in the product term: $A_{mis} \delta_{ij} B_{kr} C_{jks} = D_{mr}$,

c) The free indices m and r must occur once in each additive term. All other indices are summation indices that must occur twice per product term: $A_{msi} B_{kr} C_{jks} \delta_{ji} + D_{ml} \delta_{ur} E_{lu} = F_{mr}$

The solution is clear with regard to the name and number of indices, but the order of these indices on the symbol A is arbitrary.

Task 1.3: Transpose of a Tensor

The coefficients of the transposed tensor can be determined by evaluating both sides of Equation (1.1) and comparing them with each other. First, we represent the equation in component notation,

$$\boldsymbol{w} \cdot \boldsymbol{T}\boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{T}^T\boldsymbol{w} \qquad \Leftrightarrow \qquad w_i\boldsymbol{e}_i \cdot T_{jk}\boldsymbol{e}_j \otimes \boldsymbol{e}_k\, v_l\boldsymbol{e}_l = v_i\boldsymbol{e}_i \cdot T^T_{jk}\boldsymbol{e}_j \otimes \boldsymbol{e}_k\, w_l\boldsymbol{e}_l,$$

and simplify it as far as possible,

$$\begin{aligned} w_i\boldsymbol{e}_i \cdot T_{jk}\boldsymbol{e}_j \otimes \boldsymbol{e}_k\, v_l\boldsymbol{e}_l &= v_i\boldsymbol{e}_i \cdot T^T_{jk}\boldsymbol{e}_j \otimes \boldsymbol{e}_k\, w_l\boldsymbol{e}_l \\ w_i\boldsymbol{e}_i \cdot T_{jk}\,\delta_{kl}\,v_l\,\boldsymbol{e}_j &= v_i\boldsymbol{e}_i \cdot T^T_{jk}\,\delta_{kl}\,w_l\,\boldsymbol{e}_j \\ w_i\,T_{jk}\,v_l\,\delta_{kl}\,\delta_{ij} &= v_i\,T^T_{jk}\,w_l\,\delta_{kl}\,\delta_{ij} \\ w_i\,T_{ik}\,v_k &= v_j\,T^T_{jl}\,w_l. \end{aligned}$$

Summation indices can be renamed as desired. Since both sides must be equal, we replace l with i, and j with k on the right-hand side so that the indices of w and v match. By comparing both sides, we obtain the coefficients we are looking for:

$$\begin{aligned} &w_i\,T_{ik}\,v_k = v_k\,T^T_{ki}\,w_i \\ &w_i\,T_{ik}\,v_k = w_i\,T^T_{ki}\,v_k \qquad \Rightarrow \qquad T^T_{ki} = T_{ik} \quad \text{or} \quad T^T_{ij} = T_{ji}. \end{aligned}$$

Task 1.4: Different Notations

Symbols with one index can be represented as a column matrix (vector), symbols with two indices as a square matrix, where the first index indicates the row and the second index the column.

a) In the expression $v_i = A_{ik}w_k$, the free index i indicates that the result is a vector. The summation index k indicates the multiplication of the i th row of a matrix with the column of a vector:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{v} = \boldsymbol{A}\boldsymbol{w}.$$

b) Accordingly, the expression $C_{ik} = A_{ij}B_{jk}$ with two free indices i and k as the result represents a matrix that results from the matrix product of two matrices:

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}.$$

If it is a matrix product, the summation index (here j) must therefore be on the inside.

c) The expression $A_{ij}B_{ji}$ contains two summation indices and therefore returns a scalar as the result. The summation index j on the inside indicates a matrix multiplication, so the expression can also be interpreted as $A_{ij}B_{ji} = C_{kk}$. This shows that the sum of the main diagonal elements of a matrix is required. This can be written using the trace operator (tr):

$$\text{tr}\left(\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix}\right) \qquad \text{and} \qquad \text{tr}[\boldsymbol{AB}].$$

Alternatively, the scalar product with the unit matrix or with the unit tensor $\boldsymbol{I}$ can be used:

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{and} \qquad (\boldsymbol{AB}) \cdot \boldsymbol{I}.$$

d) The expression $u_j = A_{ik}B_{ij}v_k$ returns a vector (free index j). To convert it into matrix form, the order of the coefficients on the right-hand side must be changed. The conversion $u_j = B_{ij}A_{ik}v_k$ first shows that the product of a tensor $\boldsymbol{A}$ with a vector $\boldsymbol{v}$. To represent the operation between B_{ij} and A_{ik} as a matrix product, the order of the indices i and j must be swapped so that the summation index is on the inside. This corresponds to a transposition:

$$\begin{aligned} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} &= \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix}^T \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \\ &= \begin{bmatrix} B_{11} & B_{21} & B_{31} \\ B_{12} & B_{22} & B_{32} \\ B_{13} & B_{23} & B_{33} \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{u} = \boldsymbol{B}^T\boldsymbol{Av}. \end{aligned}$$

e) In the expression $A_{ij} - \frac{1}{3}\,\delta_{ln}\,\delta_{ij}\,\delta_{km}\,\delta_{kl}\,A_{mn}$ three of the Kronecker-deltas can be eliminated and the expression simplifies to $A_{ij} - \frac{1}{3}\,A_{mm}\,\delta_{ij}$. Now the expression can be converted directly into matrix- and symbolic notation:

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} - \tfrac{1}{3}\,(A_{11} + A_{22} + A_{33}) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{A} - \tfrac{1}{3}\,\text{tr}[\boldsymbol{A}]\,\boldsymbol{I}.$$

This is the deviatoric component of the tensor $\boldsymbol{A}$, which we will see in Task 1.6.

Task 1.5: Symmetric Tensors

The property $\boldsymbol{S} = \boldsymbol{S}^T \Rightarrow \boldsymbol{S} : \boldsymbol{T} = \boldsymbol{S} : \text{sym}(\boldsymbol{T})$ can be shown by displaying both sides of the equation in index notation. For the left-hand side, this means

$$\boldsymbol{S} : \boldsymbol{T} = (S_{ij}\boldsymbol{e}_i \otimes \boldsymbol{e}_j) : (T_{kl}\boldsymbol{e}_k \otimes \boldsymbol{e}_l) = S_{ij}\,T_{kl}\,\delta_{ik}\,\delta_{jl} = S_{ij}\,T_{ij}$$

and for the right correspondingly

$$\begin{aligned}
\boldsymbol{S} : \operatorname{sym}(\boldsymbol{T}) &= \tfrac{1}{2}\, \boldsymbol{S} : (\boldsymbol{T} + \boldsymbol{T}^T) \\
&= \tfrac{1}{2}\, (S_{ij}\, T_{ij} + S_{ij}\, T_{ji}) \\
&= \tfrac{1}{2}\, (S_{ij}\, T_{ij} + S_{ji}\, T_{ij}) && \text{(Renaming the summation indices)} \\
&= \tfrac{1}{2}\, (S_{ij}\, T_{ij} + S^T_{ij}\, T_{ij}) \\
&= \tfrac{1}{2}\, (S_{ij}\, T_{ij} + S_{ij}\, T_{ij}) && \text{(Property } S^T_{ij} = S_{ij}\text{)} \\
&= S_{ij}\, T_{ij}.
\end{aligned}$$

Task 1.6: Volumetric–Isochoric Decomposition and some Derivatives

a) The derivative of a second order tensor with respect to itself results in a fourth order tensor. Each component of the strain tensor is differentiated with respect to each component of the strain tensor itself and this results in a fourth order tensor with $9 \cdot 9 = 81$ new components. In coefficient notation:

$$\frac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\varepsilon}} = \frac{\partial (\varepsilon_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j)}{\partial (\varepsilon_{kl} \boldsymbol{e}_k \otimes \boldsymbol{e}_l)} \quad \widehat{=} \quad \frac{\partial \varepsilon_{ij}}{\partial \varepsilon_{kl}} = \delta_{ik}\, \delta_{jl} = \mathbb{I}_{ijkl} \qquad \Rightarrow \qquad \frac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\varepsilon}} = \mathbb{I}$$

b) The derivative of the trace of a second order tensor, i.e. a scalar, with respect to the tensor itself results in a second order tensor and can be determined in the following three ways:

$$\frac{\partial \operatorname{tr}[\boldsymbol{\varepsilon}]}{\partial\, \boldsymbol{\varepsilon}} \quad \widehat{=} \quad \frac{\partial\, \varepsilon_{kk}}{\partial\, \varepsilon_{ij}} = \delta_{ki}\, \delta_{kj} = \delta_{ij} \qquad \Rightarrow \qquad \frac{\partial \operatorname{tr}[\boldsymbol{\varepsilon}]}{\partial\, \boldsymbol{\varepsilon}} = \boldsymbol{I}$$

$$\frac{\partial \operatorname{tr}[\boldsymbol{\varepsilon}]}{\partial\, \boldsymbol{\varepsilon}} = \frac{\partial\, (\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33})}{\partial\, \boldsymbol{\varepsilon}} \quad \widehat{=} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \widehat{=} \quad \boldsymbol{I}$$

$$\begin{aligned}
\frac{\partial\, (\boldsymbol{I} : \boldsymbol{\varepsilon})}{\partial\, \boldsymbol{\varepsilon}} &= \frac{\partial\, \boldsymbol{I}}{\partial\, \boldsymbol{\varepsilon}} : \boldsymbol{\varepsilon} + \boldsymbol{I} : \frac{\partial\, \boldsymbol{\varepsilon}}{\partial\, \boldsymbol{\varepsilon}} \\
&= \boldsymbol{I} : \mathbb{I} \\
&= \delta_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j : \delta_{km}\, \delta_{ln} \boldsymbol{e}_k \otimes \boldsymbol{e}_l \otimes \boldsymbol{e}_m \otimes \boldsymbol{e}_n \\
&= \delta_{ij}\, \delta_{km}\, \delta_{ln}\, \delta_{ik}\, \delta_{jl}\, \boldsymbol{e}_m \otimes \boldsymbol{e}_n \\
&= \delta_{mn} \boldsymbol{e}_m \otimes \boldsymbol{e}_n \\
&= \boldsymbol{I}
\end{aligned}$$

c) The derivative of the deviator can be easily determined by using the definition of the deviator:

$$\frac{\partial \operatorname{dev}[\boldsymbol{\varepsilon}]}{\partial\, \boldsymbol{\varepsilon}} = \frac{\partial\, (\boldsymbol{\varepsilon} - \frac{1}{3} \operatorname{tr}[\boldsymbol{\varepsilon}]\, \boldsymbol{I})}{\partial\, \boldsymbol{\varepsilon}} = \mathbb{I} - \tfrac{1}{3}\, \frac{\partial \operatorname{tr}[\boldsymbol{\varepsilon}]}{\partial\, \boldsymbol{\varepsilon}} \otimes \boldsymbol{I} = \mathbb{I} - \tfrac{1}{3} \boldsymbol{I} \otimes \boldsymbol{I}$$

Please note that the result is a fourth order tensor. The second part of the right-hand side must therefore be a dyadic product of two second order Identity tensors. The result is a special projection tensor $\mathbb{P} = \mathbb{I} - \frac{1}{3}\boldsymbol{I} \otimes \boldsymbol{I}$, which maps a second order tensor to its deviatoric part.

d) The derivative of the norm of the deviator of $\boldsymbol{\varepsilon}$ with respect to $\boldsymbol{\varepsilon}$ results in a second order tensor. We apply the chain rule to evaluate it:

$$\begin{aligned}
\frac{\partial \,||\operatorname{dev}\boldsymbol{\varepsilon}||}{\partial\,\boldsymbol{\varepsilon}} &= \frac{\partial \,||\operatorname{dev}\boldsymbol{\varepsilon}||}{\partial \operatorname{dev}[\boldsymbol{\varepsilon}]} : \frac{\partial \operatorname{dev}[\boldsymbol{\varepsilon}]}{\partial\,\boldsymbol{\varepsilon}} \\
&= \frac{\partial \sqrt{(\operatorname{dev}\boldsymbol{\varepsilon})_{11}^2 + (\operatorname{dev}\boldsymbol{\varepsilon})_{12}^2 + \ldots + (\operatorname{dev}\boldsymbol{\varepsilon})_{33}^2}}{\partial\ \operatorname{dev}[\boldsymbol{\varepsilon}]} : \mathbb{P} \\
&= \frac{1}{2||\operatorname{dev}\boldsymbol{\varepsilon}||}\, 2\operatorname{dev}[\boldsymbol{\varepsilon}] : \mathbb{P} \\
&= \frac{\operatorname{dev}[\boldsymbol{\varepsilon}]}{||\operatorname{dev}\boldsymbol{\varepsilon}||}.
\end{aligned}$$

Since applying the projection tensor $\mathbb{P}$ to $\operatorname{dev}[\boldsymbol{\varepsilon}]$ has no effect, the expression to the left of the projection tensor remains unchanged at the end.

Task 1.7: Matrix and Voigt Notations of Tensors

From Equation (1.8), we observe that lines 7 to 9 of the system of equations can be omitted, since the stress components σ_{21}, σ_{32} and σ_{31} are already known from lines 4 to 6 due to the symmetry of $\boldsymbol{\sigma}$. The system of equations thus reduces to

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113} & \mathbb{C}_{1121} & \mathbb{C}_{1132} & \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213} & \mathbb{C}_{2221} & \mathbb{C}_{2232} & \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313} & \mathbb{C}_{3321} & \mathbb{C}_{3332} & \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213} & \mathbb{C}_{1221} & \mathbb{C}_{1232} & \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313} & \mathbb{C}_{2321} & \mathbb{C}_{2332} & \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313} & \mathbb{C}_{1321} & \mathbb{C}_{1332} & \mathbb{C}_{1331} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{32} \\ \varepsilon_{31} \end{bmatrix}.$$

Since we know that $\boldsymbol{\varepsilon}$ is also symmetrical, columns 7 to 9 can also be eliminated. To do this, add the colored entries of $\mathbb{C}$ at the corresponding position three columns to the left. For example, because $\varepsilon_{21} = \varepsilon_{12}$, row 7 of the column vector and column 7 of the matrix can be omitted if the latter is added to column 4 of the matrix. The system of equations then becomes

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} + \mathbb{C}_{1121} & \mathbb{C}_{1123} + \mathbb{C}_{1132} & \mathbb{C}_{1113} + \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} + \mathbb{C}_{2221} & \mathbb{C}_{2223} + \mathbb{C}_{2232} & \mathbb{C}_{2213} + \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} + \mathbb{C}_{3321} & \mathbb{C}_{3323} + \mathbb{C}_{3332} & \mathbb{C}_{3313} + \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} + \mathbb{C}_{1221} & \mathbb{C}_{1223} + \mathbb{C}_{1232} & \mathbb{C}_{1213} + \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} + \mathbb{C}_{2321} & \mathbb{C}_{2323} + \mathbb{C}_{2332} & \mathbb{C}_{2313} + \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} + \mathbb{C}_{1321} & \mathbb{C}_{1323} + \mathbb{C}_{1332} & \mathbb{C}_{1313} + \mathbb{C}_{1331} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \end{bmatrix}.$$

By now taking into account the symmetries of $\mathbb{C}$ from Equation (1.9) the system of equations can be simplified

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{23}\\ \sigma_{13}\end{bmatrix} = \begin{bmatrix}\mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & 2\,\mathbb{C}_{1112} & 2\,\mathbb{C}_{1123} & 2\,\mathbb{C}_{1113}\\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & 2\,\mathbb{C}_{2212} & 2\,\mathbb{C}_{2223} & 2\,\mathbb{C}_{2213}\\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & 2\,\mathbb{C}_{3312} & 2\,\mathbb{C}_{3323} & 2\,\mathbb{C}_{3313}\\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & 2\,\mathbb{C}_{1212} & 2\,\mathbb{C}_{1223} & 2\,\mathbb{C}_{1213}\\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & 2\,\mathbb{C}_{2312} & 2\,\mathbb{C}_{2323} & 2\,\mathbb{C}_{2313}\\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & 2\,\mathbb{C}_{1312} & 2\,\mathbb{C}_{1323} & 2\,\mathbb{C}_{1313}\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ \varepsilon_{12}\\ \varepsilon_{23}\\ \varepsilon_{13}\end{bmatrix},$$

and the factor 2 can be shifted to the strains to obtain

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{23}\\ \sigma_{13}\end{bmatrix} = \begin{bmatrix}\mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113}\\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213}\\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313}\\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213}\\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313}\\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313}\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ 2\,\varepsilon_{12}\\ 2\,\varepsilon_{23}\\ 2\,\varepsilon_{13}\end{bmatrix},$$

The remaining $6\,\mathrm{x}\,6$ matrix turns out to be symmetrical due to Equation (1.10) and can therefore be represented by 21 remaining coefficients:

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{23}\\ \sigma_{13}\end{bmatrix} = \begin{bmatrix}\mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113}\\ & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213}\\ & & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313}\\ & & & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213}\\ & \text{sym.} & & & \mathbb{C}_{2323} & \mathbb{C}_{2313}\\ & & & & & \mathbb{C}_{1313}\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ 2\,\varepsilon_{12}\\ 2\,\varepsilon_{23}\\ 2\,\varepsilon_{13}\end{bmatrix}.$$

In order to represent the projection tensor $\mathbb{P} = \mathbb{I} - \frac{1}{3}\,\boldsymbol{I} \otimes \boldsymbol{I}$ in Voigt notation, we need both the fourth order unit tensor and the term $\boldsymbol{I} \otimes \boldsymbol{I}$ in Voigt notation.

Since the contraction of the strain tensor with the Identity tensor should result in the strain tensor itself i. e. $\mathbb{I} : \boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}$, this must also be true in the Voigt notation $\underline{\mathbb{I}}\,\underline{\boldsymbol{\varepsilon}} = \underline{\boldsymbol{\varepsilon}}$. Constructing the fourth order Identity tensor in Voigt notation to fit this equation results in

$$\underline{\mathbb{I}} = \begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0\\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0\\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}\end{bmatrix}$$

when $\boldsymbol{\varepsilon}$ in Voigt notation is given as

$$\underline{\boldsymbol{\varepsilon}} = \begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ 2\,\varepsilon_{12}\\ 2\,\varepsilon_{23}\\ 2\,\varepsilon_{13}\end{bmatrix}$$

Now, the term $\boldsymbol{I} \otimes \boldsymbol{I}$ can be evaluated using the second order unit tensor in Voigt notation $\underline{\boldsymbol{I}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}^T$, and as the result of the matrix multiplication $\underline{\boldsymbol{I}}\,\underline{\boldsymbol{I}}^T$, we obtain

$$\underline{\boldsymbol{I}}\,\underline{\boldsymbol{I}}^T = \underline{\boldsymbol{I}} \otimes \underline{\boldsymbol{I}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Thus, the matrix representation of the projection tensor follows from the above results

$$\underline{\mathbb{P}} = \begin{bmatrix} +\frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 & 0 \\ -\frac{1}{3} & +\frac{2}{3} & -\frac{1}{3} & 0 & 0 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & +\frac{2}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} .$$